CVPR 2025 · HIGHLIGHT

AIpparel: A Multimodal Foundation Model for Digital Garments

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Fig. 1

AGENDA

Roadmap

From motivation to a multimodal foundation model for garments

01 **Title** — Paper identity & venue

02 **Outline** — Roadmap of the talk

03 **Introduction / Background** — Why digital garments matter

04 **Limitations** — Gaps in single-modal methods

05 **Overview** — Core idea & contributions

06 **Benchmark / Dataset** — GCD-MM construction

07 **Key Algorithm** — Drawing-command tokenization — PEAK

08 **Dataset & Training** — Data stats & objective

09 **Experimental Setup** — Tasks, baselines, metrics

10 **Results & Analysis** — SOTA quantitative gains — PEAK

11 **Visual Analysis & Editing** — Qualitative editing

12 **Takeaways & Limitations** — What we learned

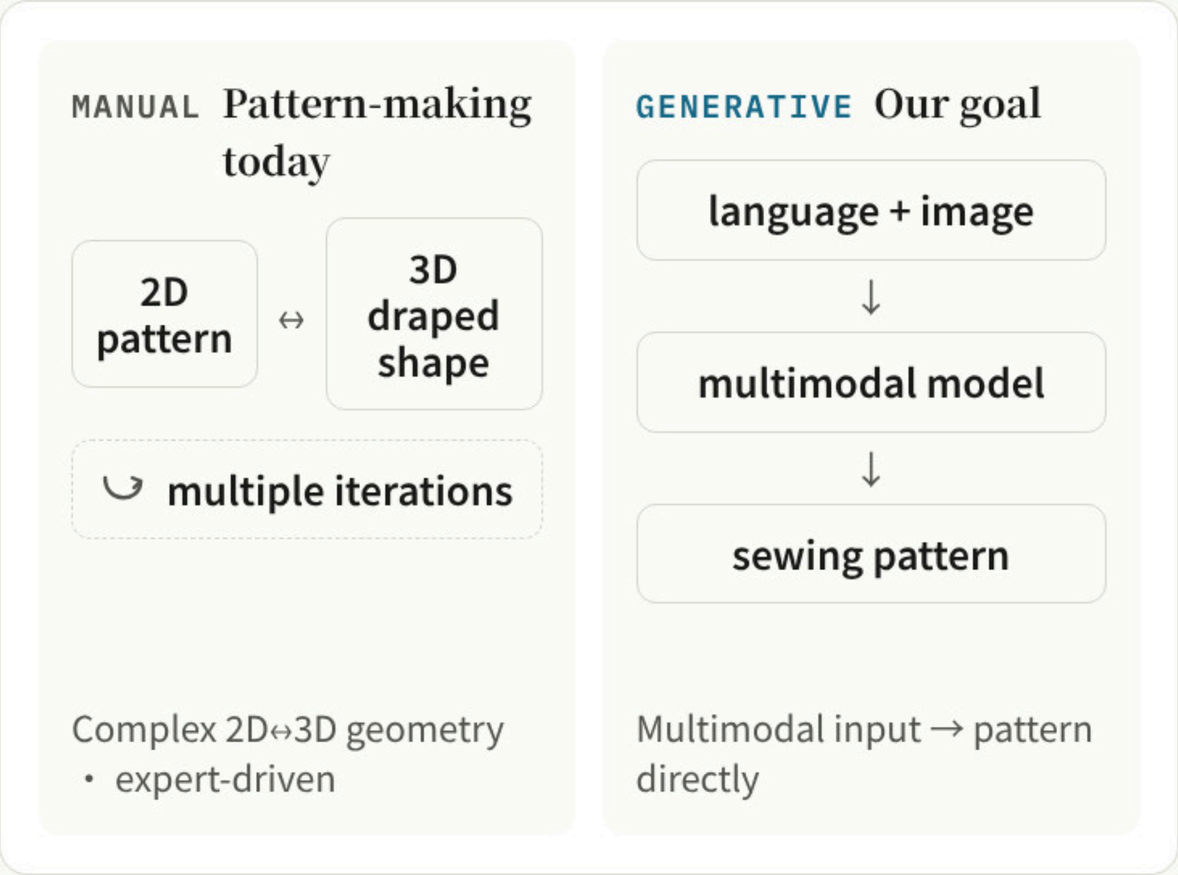
13 **Conclusion** — First multimodal foundation model

- Two peaks anchor the talk: **tokenization** (§7) and **SOTA results** (§10).

INTRODUCTION

Digital garment creation is essential — but manual

Pattern-making is a time-consuming, iteration-heavy process



SEWING PATTERN

A set of flat 2D panels with standardized assembly instructions that form a complete 3D garment.

- Clothing is central to **identity** and personal style.
- Pattern-making is the key stage — and it's **manual**.
- Current SOTA methods are **single-modal**: 3D points, images, or language.
- We need a scalable model that handles **language + image** together.

LIMITATIONS

Existing methods are single-modal and limited

Task-specific models, scarce multimodal data, simple geometries

01 **Single-modality focus**

Models are task-specific; adapting to combined inputs requires redesigning the network.

02 **Data scarcity**

No large multimodal sewing-pattern dataset with text, image, and editing pairs.

03 **Complexity limits**

Prior models restricted to simple garments with predefined parameters.



Fig. 1 AIpparel generates a complex knee-length jumpsuit from text + image, directly 3D-simulatable.

OVERVIEW

A scalable recipe: fine-tune an LMM on a curated dataset

Three contributions — dataset, tokenization, multimodal capability

01 GCD-MM

Large-scale multimodal dataset

First sewing-pattern dataset with text, images and editing instructions — extending GCD with multimodal annotations.

120k+ unique garments

02 TOKENIZATION

Efficient, learning-friendly tokens

A drawing-command tokenizer that fits complex patterns within an LMM's limited context length.

~100× fewer tokens

03 MULTIMODAL

New multimodal capability

First multimodal foundation model for patterns — takes language, images and patterns as input.

Language- instructed editing

CORE IDEA | Fine-tune a large multimodal model on a curated dataset of **120,000+** unique garments — each annotated with text, images and sewing patterns.

METHOD · DATASET

GCD-MM: a large-scale multimodal sewing-pattern dataset

Extend GCD with text, images, and editing pairs

01

Extend GCD

Build on GarmentCodeData (GCD) [30], the largest existing sewing-pattern dataset.

02

Rule-based + GPT-4o annotation

A rule-based algorithm extracts key features; GPT-4o writes accurate natural-language descriptions, reducing hallucination.

03

Editing pairs

Use GarmentCode's programming abstraction to create paired patterns with text editing instructions, e.g. "make the skirt longer".

120k unique garments with text, image & edit annotations — **10× larger** than prior SOTA generative datasets.

Table 1 · Modalities of sewing-pattern datasets

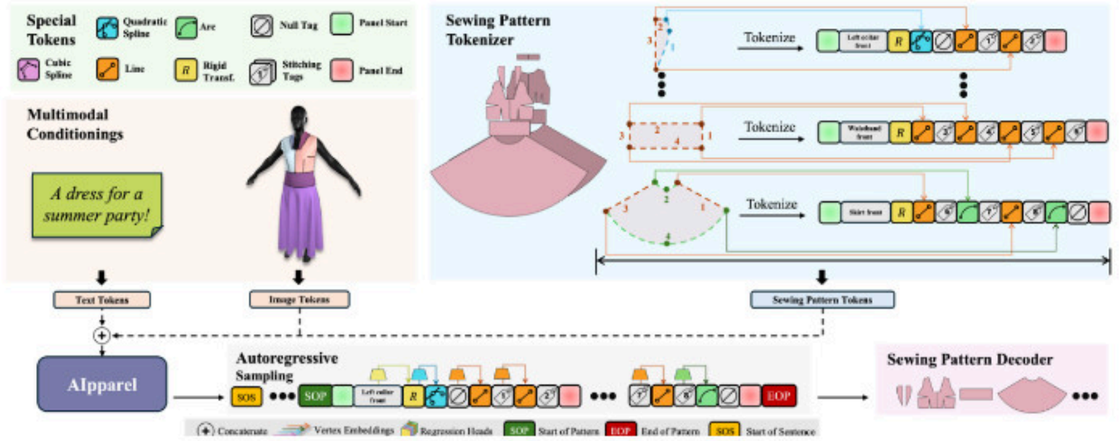
DATASET	TOTAL	TEXT	IMAGE	EDITS
Wang et al. [63]	8k	✗	✓	✓
Korosteleva & Lee [27]	23.5k	✗	✓	✗
SewFactory [36]	19.1k	✗	✓	✗
DressCode [21]	20.3k	✓	✗	✗
GCD [30]	130k	✗	✓	✗
GCD-MM (Ours)	120k	✓	✓	✓

Only GCD-MM carries all three modalities together.

KEY ALGORITHM

Tokenizing sewing patterns as drawing commands

Special tokens + ~100x reduction in context length



Multimodal conditionings and sewing patterns are tokenized into drawing-command sequences, autoregressively sampled, and decoded into simulatable patterns.

Fig. 2 · Sec 3.2

A pattern = 2D panels + stitching info.

Special tokens

<SoG>
Garment start

<EoG>
Garment end

<SoP>
Panel start

<EoP>
Panel end

Panel token

<SoP> [name] <R> <L>/<Q>/<B>/<A> <t_s>…<EoP>

line <L> • quadratic <Q> • cubic <B> • arc <A>

~250
avg tokens • max 838

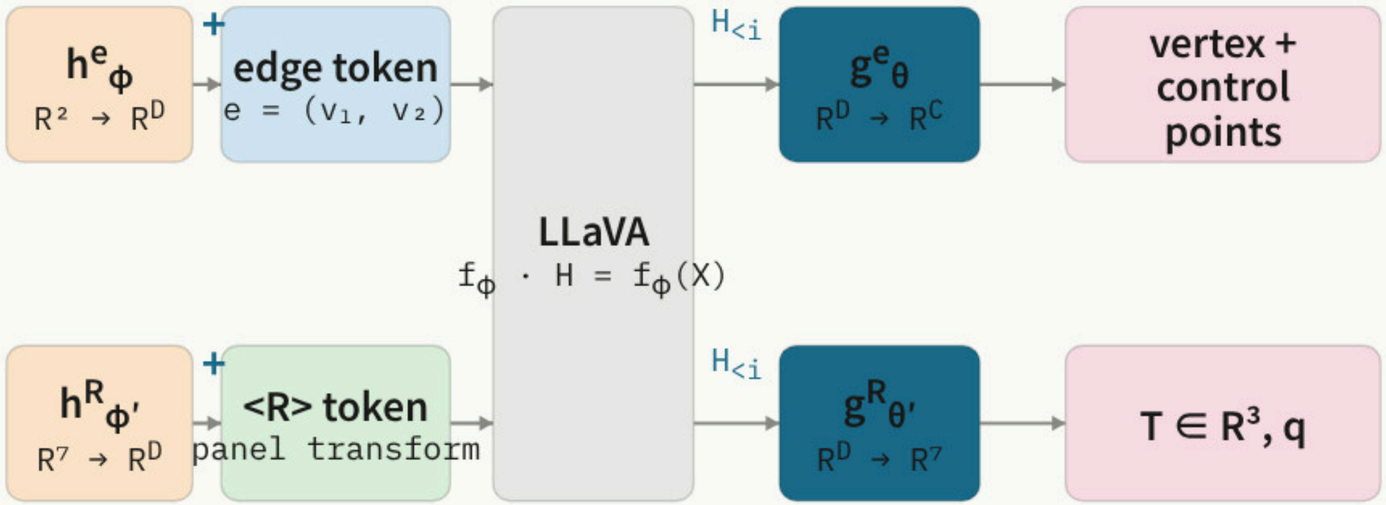
>30k
DressCode

≈ 100× shorter context

KEY ALGORITHM

Regression heads + positional embeddings for continuous parameters

Hybrid discrete tokens and regressed continuous values



Sec. 3.2

Continuous parameters, handled by regression

Vertex positions, edge control points, and panel rigid transforms are regressed, not quantized.

Regression heads

Small MLPs map the hidden embedding to continuous values: $g^e_\theta : R^D \to R^C$ (vertices, control points), $g^R_{\theta'} : R^D \to R^7$ ($T \in R^3$, q).

Positional embeddings

Values are injected back into token embeddings: $h^e_\phi : R^2 \to R^D$ adds v_2 to the edge token; $h^R_{\phi'} : R^7 \to R^D$ adds the transform to the <R> token.

Why it matters

Avoids quantization error and keeps context compact.

TRAINING

Dataset statistics and training objective

120k+ garments; CE + L2 on a frozen-encoder LLaVA 1.5-7B

120k+

unique garments

multimodal annotations — text • image • sewing pattern

Training objective

Eq. 2

$$L = \text{CE}_{\text{discrete tokens}} + \lambda_{e'} \left\| g^{e}_{\theta} \circ f_{\phi}(X) - v^{2}_{e'} \right\| + \lambda_{R'} \left\| g^{R}_{\theta'} \circ f_{\phi}(X) - R' \right\|$$

CE — discrete tokens

L2 — continuous parameters

λ — weighting terms

GCD-MM split

random 90 : 5 : 5

train 90% | val 5% | test 5%

single model for all GCD-MM results

Model & hardware

LLaVA 1.5-7B

frozen — vision encoder + projection

fine-tuned — language model f_{ϕ}

fine-tuned — regression heads g^{e}_{θ} • $g^{R}_{\theta'}$

fine-tuned — positional-embedding projections h^{e}_{ϕ} • $h^{R}_{\phi'}$

EXPERIMENTAL SETUP

Tasks, baselines, and metrics

Three tasks, two baselines, four reconstruction metrics

Metric	Direction	Unit	What it measures
Panel L2	↓	cm	Avg vertex L2 distance between predicted and GT panels
#Panel Accuracy	↑	%	Patterns with the correctly predicted number of panels
#Edge Accuracy	↑	%	Correctly predicted edges in each correctly predicted panel
#Stitch Accuracy	↑	%	Predicted stitches compared to ground truth

Accuracy = #Panel Acc × #Edge Acc All L2 metrics in cm except rotation

TASKS

- Image-to-Garment
- Text-to-Garment
- Language-instructed Editing

BASELINES

- SewFormer Liu et al. 2023b
- DressCode He et al. 2024

AIpparel outperforms SewFormer-FT by a large margin

#Stitch Accuracy 77.2% vs 2.8% on GCD-MM

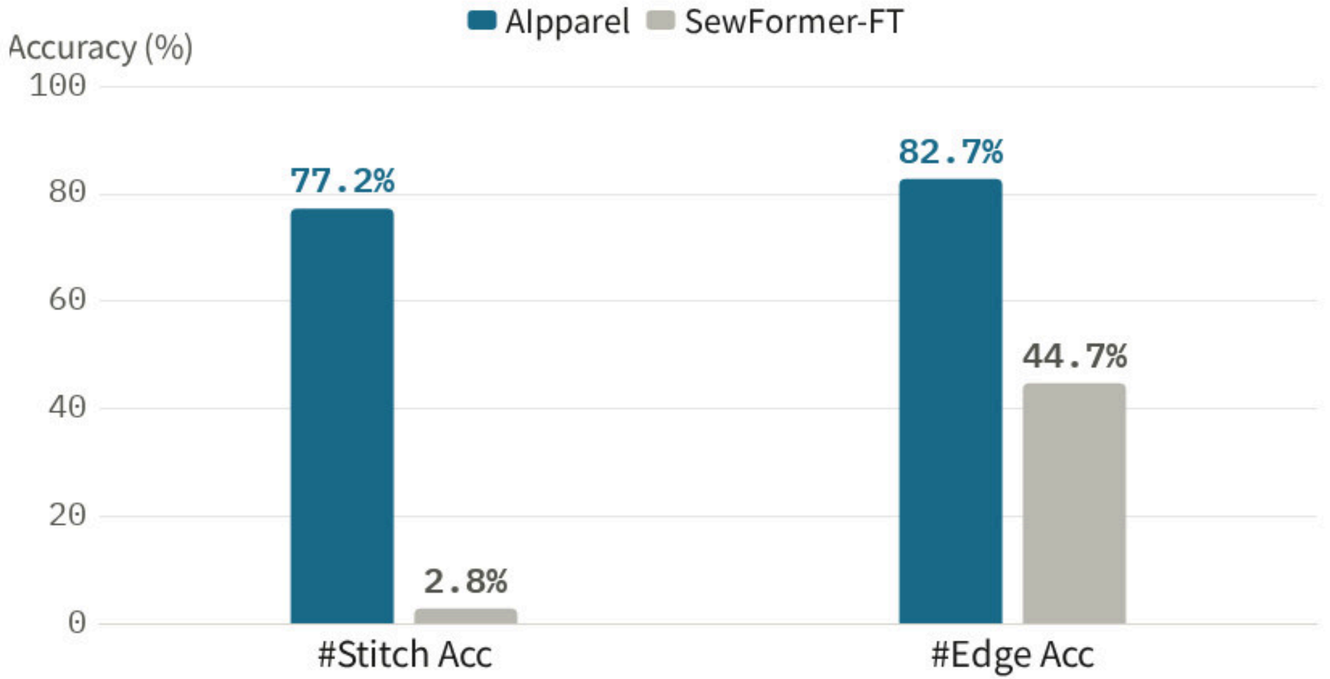


Table 2 · GCD-MM M1 p.8142

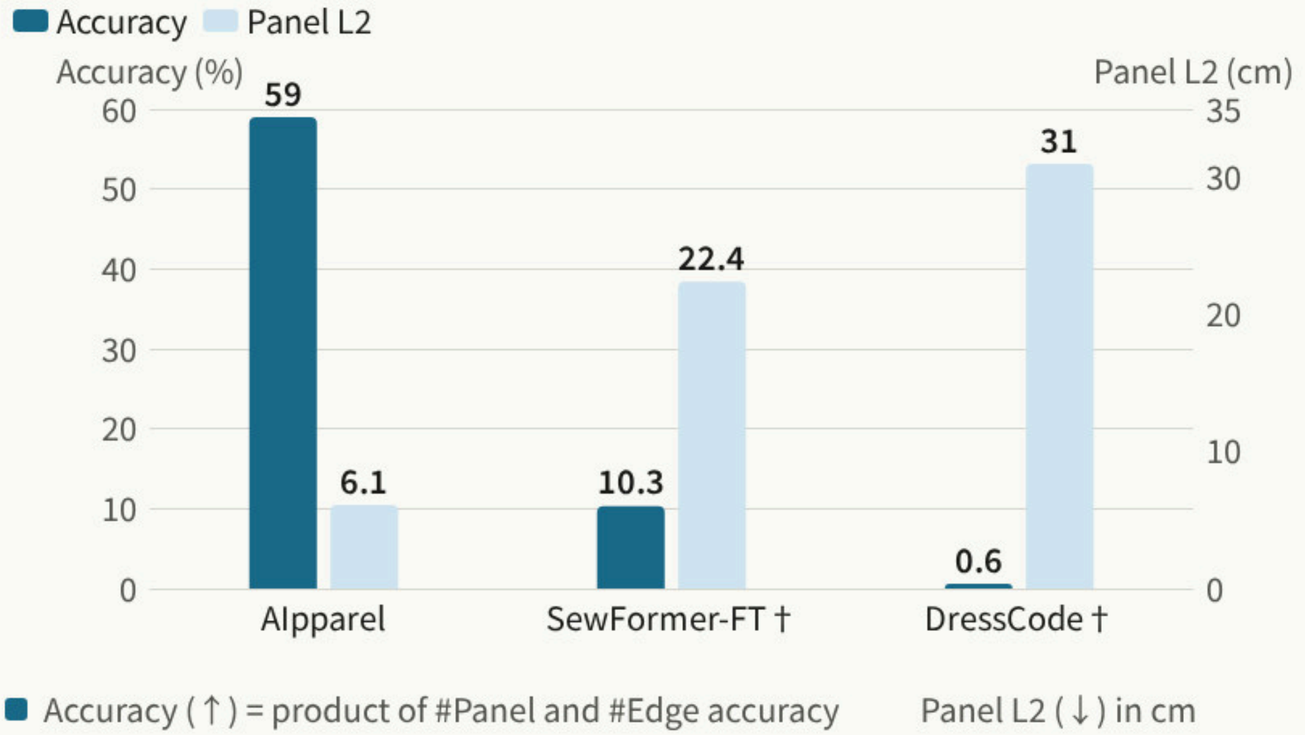
METRIC	AIPPAREL	SEWFORMER-FT
Panel L2 ↓	5.4	12.3
#Panel Acc	85.2	79.4
#Edge Acc	**82.7**	**44.7**
#Stitch Acc	**77.2**	**2.8**
Rot L2 ↓	.020	.040
Transl L2 ↓	2.7	4.5

Direct multimodal generation beats augmented baselines

Accuracy 59.0 vs 10.3 (SewFormer-FT †) and 0.6 (DressCode †)

Multimodal generation on 100 garments

5 input types • 20 test samples each



Accuracy (↑) = product of #Panel and #Edge accuracy

Panel L2 (↓) in cm

Table 3 · Multimodal prediction

Table 3

METHOD	ACCURACY (↑)	PANEL L2 (↓)
AIpparel	59.0	6.1
SewFormer-FT †	10.3	22.4
DressCode †	0.6	31.0

† Baselines augmented with GPT-4o / DALL-E to translate multimodal inputs into their single-modal domains. AIpparel handles all multimodal tasks directly, with no external modules.

Language-instructed editing with minimal style drift

Accuracy 83.4 vs 9.5 (SewFormer*) and 37.0 (DressCode*)

Sewing pattern editing · Table 4

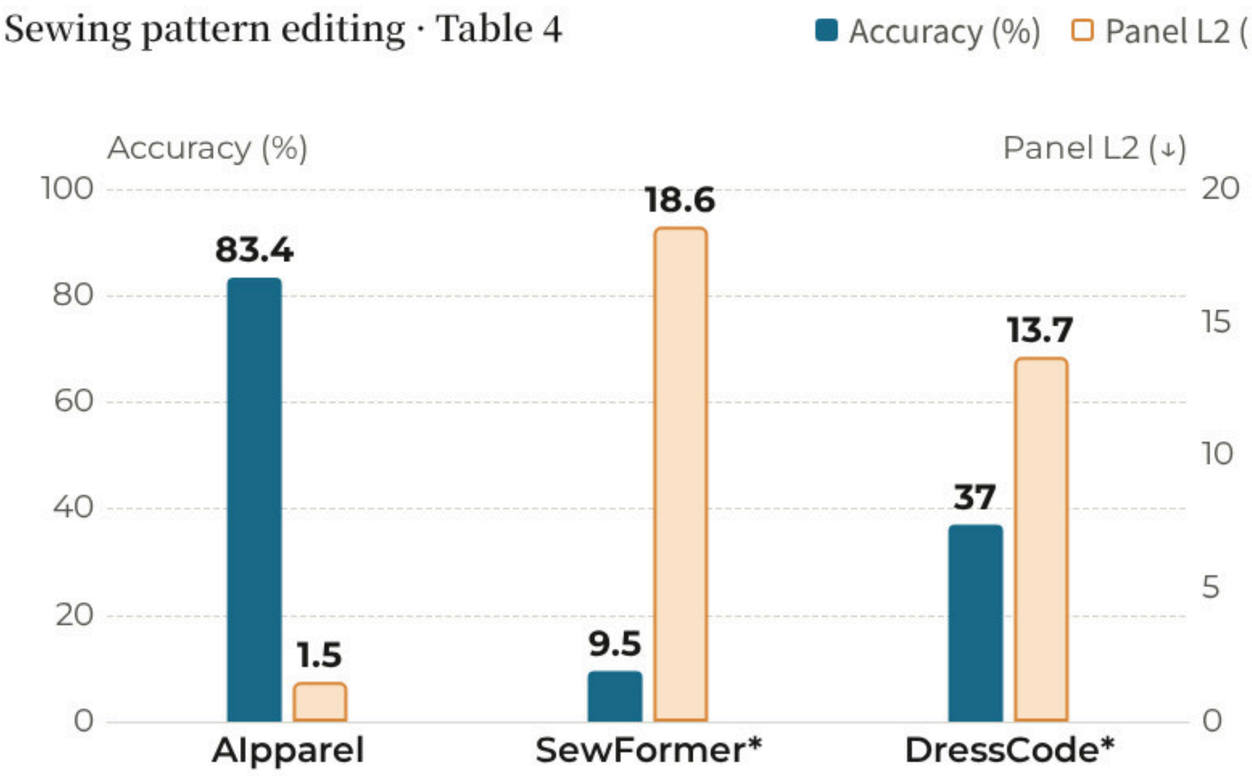


Table 4 · Sewing Pattern Editing

Higher Accuracy and lower Panel L2 are better

METHOD	ACCURACY (↑)	PANEL L2 (↓)
AIpparel	83.4	1.5
SewFormer*	9.5	18.6
DressCode*	37.0	13.7

AIpparel takes the pattern and instruction directly, making the minimal edit without drifting from the original style.

ABLATION

Tokenizer and objective both matter — and are 25x faster

Accuracy 85.0 vs 38.4 (DressCode); 52.2s → 2.1s

Accuracy vs. inference time

Text-to-garment • DressCode dataset • single A6000

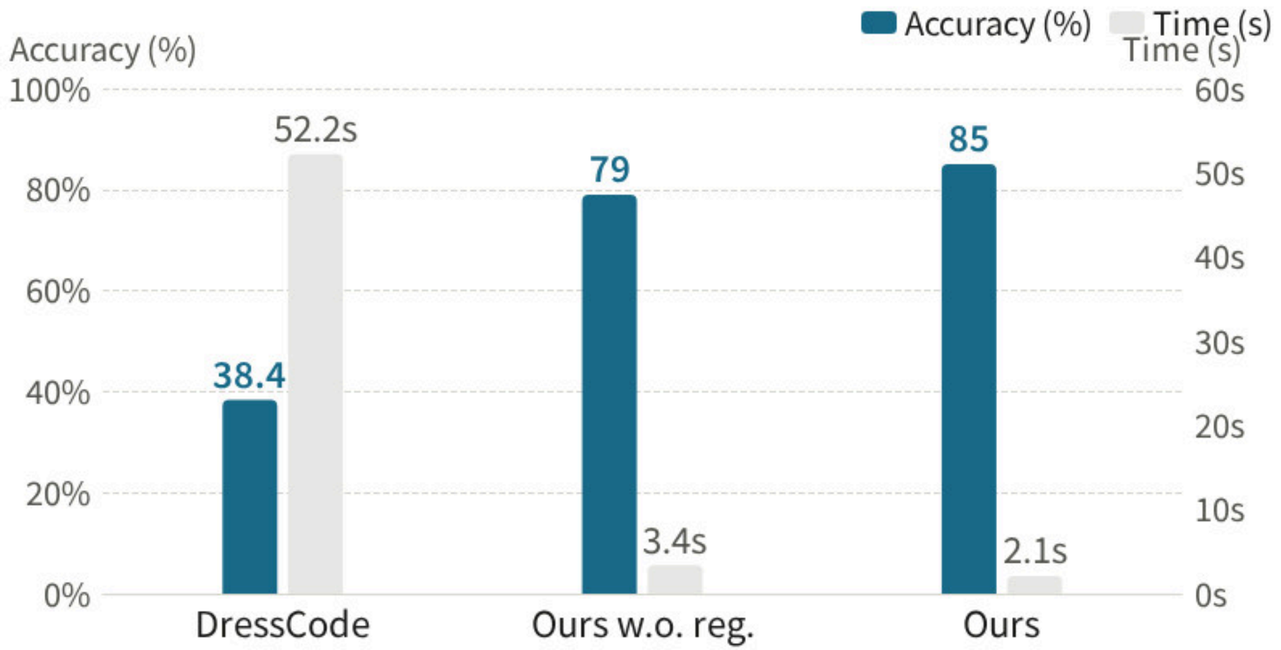


Table 5

25×
faster inference
52.2s → 2.1s per garment

Table 5 · Ablation

METHOD	ACC. ↑	PANEL L2 ↓	TIME ↓
DressCode	38.4	22.4	52.2s
Ours w.o. reg.	79.0	7.2	3.4s
Ours	**85.0**	**6.1**	**2.1s**

Same backbone; only tokenizer & training objective change.

QUALITATIVE ANALYSIS

Precise edits and fine-grained reconstruction

Hoods, longer skirts, waistbands, and sleeve cuffs

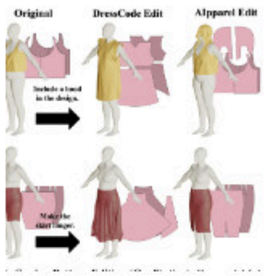


Fig. 5 Editing precision



Fig. 4 Common-sense reasoning

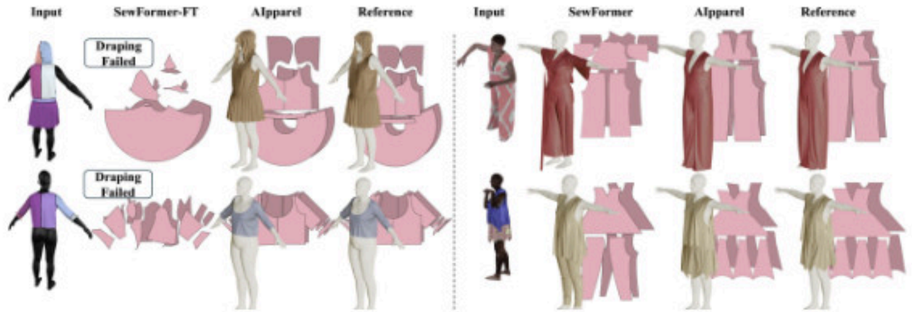


Fig. 3 Reconstruction detail

01 Editing precision

Follows “include a hood” and “make the skirt longer” while **preserving the original style**.

02 Reconstruction detail

Recovers small panels baselines miss — the **waistband** and **sleeve cuffs**.

03 Common-sense reasoning

For a “semi-formal garden party”, AIpparel outputs a **godet skirt**; DressCode † returns an unrelated **mini-skirt**.

TAKEAWAYS & LIMITATIONS

Efficient tokenization is the key — with open limits

Multimodal training enables intuitive design; synthetic-data and non-manifold limits remain

TAKEAWAYS

`pattern → tokenizer → tokens → LMM → garment`

01 Efficient tokenization is critical

A concise token scheme is what makes it feasible to scale LMMs to complex garment geometries.

02 Multimodal training enables intuitive design

Language and image inputs let users generate and edit sewing patterns directly.

LIMITATIONS

A Procedural data

Relies on synthetically generated data; a gap to real 3D scans remains open.

B Non-manifold structures

Representation is constrained to manifold surfaces — pockets need non-manifold support.

CONCLUSION

A scalable recipe for multimodal garment generation

First multimodal foundation model for sewing patterns; releasing GCD-MM



From a garment description to a drapeable, production-ready pattern.